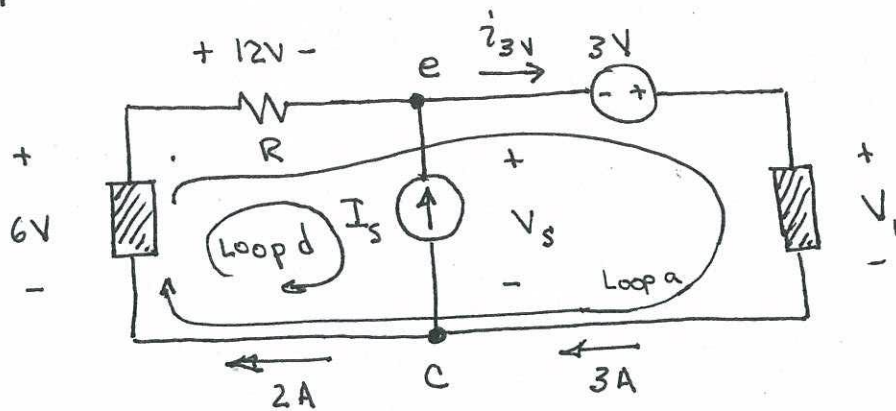


1.1



a) KVL, Loop a: $-6V + 12V - 3V + V_1 = 0 \Rightarrow \boxed{V_1 = -3V}$

b) current through R is 2A (from KCL), so from Ohm's Law:

$$(2A)(R) = 12V \Rightarrow \boxed{R = 6\Omega}$$

c) KCL at node "c"

$$2A + I_s = 3A \Rightarrow \boxed{I_s = 1A}$$

d) KVL, Loop d:

$$-6V + 12V + V_s = 0 \Rightarrow V_s = -6V$$

Note: current enters positive voltage node.

$$P = i v \Rightarrow P = (6V)(1A)$$

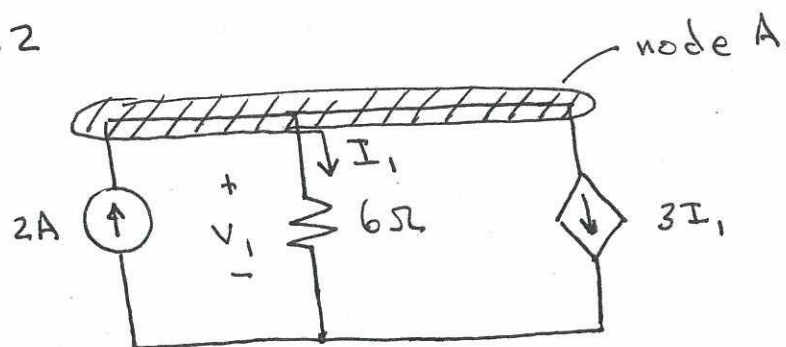
$$\boxed{P = 6W \text{ (absorbed)}}$$

e) KCL, node e

$$i_{3V} = 2A + 1A = 3A \text{ (enters negative voltage node)}$$

$$P = i v = (-3A)(3V) \Rightarrow \boxed{P = -9W \text{ (generated)}}$$

1.2



KCL, node A:

$$2A = I_1 + 3I_1$$

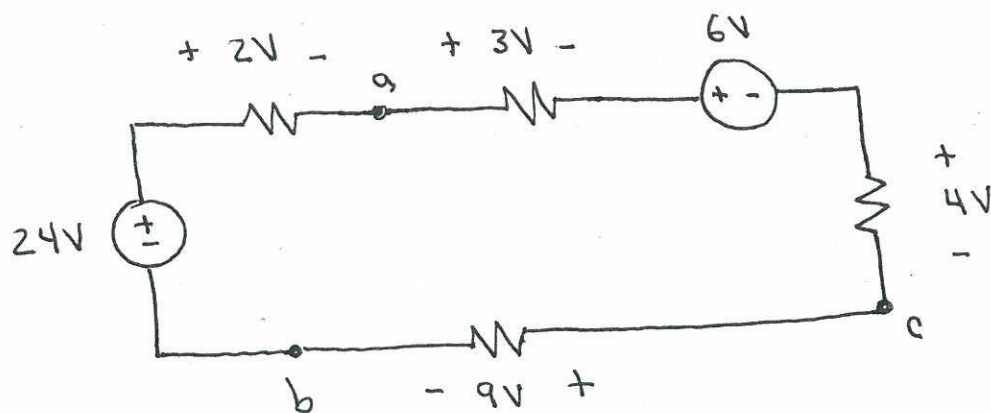
$$2A = 4I_1 \Rightarrow I_1 = 0.5A$$

Ohm's Law:

$$V_1 = (6\Omega) I_1 = (6\Omega)(0.5A)$$

$$V_1 = 3V$$

1.3



$$\text{KVL: } -24\text{V} + 2\text{V} + v_{ab} = 0$$

$$v_{ab} = 22\text{V}$$

$$\text{KVL: } -24\text{V} + 2\text{V} + v_{ac} + 9\text{V} = 0$$

$$v_{ac} = 13\text{V}$$

By inspection,

$$v_{cb} = 9\text{V}$$

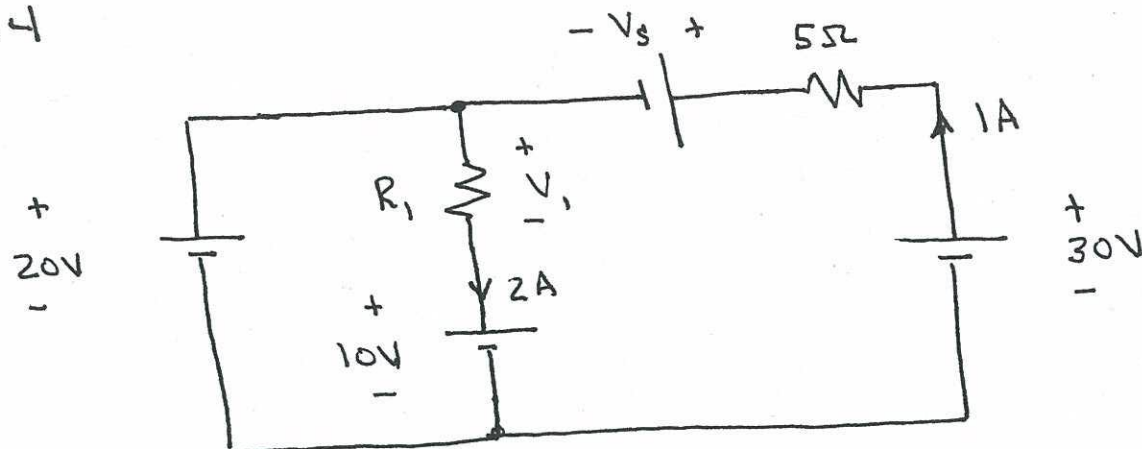
basically given,
though KVL gives

$$v_{bc} + 9\text{V} = 0;$$

$$\text{since } v_{cb} = -v_{bc}$$

$$\Rightarrow \underline{\underline{v_{cb} = 9\text{V}}}$$

1.4



(a) KVL around outer loop:

$$-20V - V_s - \underbrace{(5\Omega)(1A)}_{\text{Ohm's Law}} + 30V = 0 \Rightarrow \boxed{V_s = 5V}$$

(b) KVL around leftmost loop:

$$-20V + V_1 + 10V = 0 \Rightarrow V_1 = 10V$$

Ohm's Law:

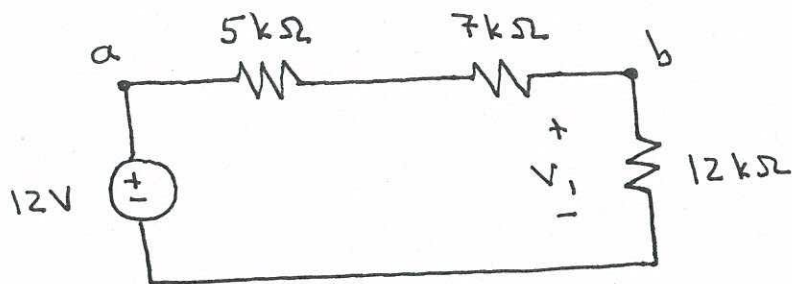
$$V_1 = R_1(2A)$$

$$\rightarrow R_1 = \frac{10V}{2A} \Rightarrow \boxed{R_1 = 5\Omega}$$

(c) $P = i v = (2A)(10V) \Rightarrow \boxed{P = 20W \text{ (absorbed)}}$

current enters positive voltage node $\Rightarrow$ power absorbed

1.5



Note: KCL, applied at any node in the circuit, tells us that the current (i) is the same in all circuit elements.

KVL: $-12V + v_{ab} + V_1 = 0$

ohm's law:

$$v_{ab} = 5k\Omega \cdot i + 7k\Omega \cdot i = (12 \times 10^3 \Omega) i$$

$$V_1 = 12k\Omega \cdot i = (12 \times 10^3 \Omega) i$$

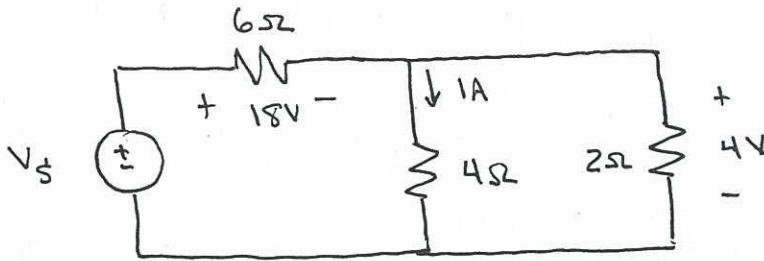
$$0 = -12V + (12 \times 10^3) i + (12 \times 10^3) i$$

$$12V = (24 \times 10^3) i \Rightarrow i = 0.5 \text{ mA}$$

$$v_{ab} = (12 \times 10^3) i = (12 \times 10^3)(0.5 \times 10^{-3}) \Rightarrow \boxed{v_{ab} = 6V}$$

$$V_1 = (12 \times 10^3) i \Rightarrow \boxed{V_1 = 6V}$$

1.6



$$(a) \quad P = \frac{v^2}{R} = \frac{(18V)^2}{6\Omega} = \boxed{54W}$$

$$(b) \quad v = iR = (1A)(4\Omega) = \boxed{4V}$$

$$(c) \quad i = \frac{v}{R} = \frac{4V}{2\Omega} = \boxed{2A}$$

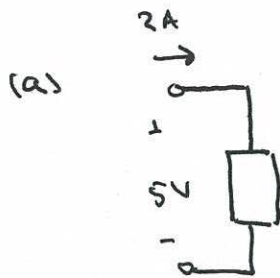
$$(a) \quad P = i v = (2A)(5V) = \boxed{10W} \text{ (absorbed - current enters positive voltage node)}$$

$$(b) \quad P = (-1A)(3V) = \boxed{-3W} \text{ (generated - current enters negative voltage node)}$$

$$(c) \quad P = (3A)(10V) = \boxed{30W} \text{ (absorbed - current leaves negative voltage node; same as entering positive voltage node)}$$

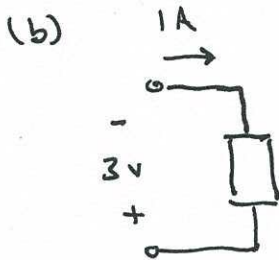
$$(d) \quad P = (-2A)(2V) = \boxed{-4W} \text{ (generated; negative current enters positive voltage node)}$$

1.7



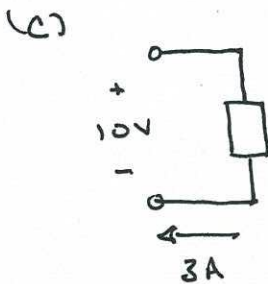
$$P = (2A)(5V) = \underline{10W} \text{ (absorbed)}$$

↑ Agree with passive sign convention



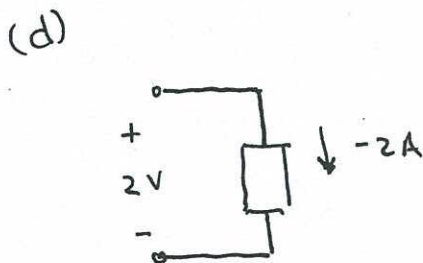
$$P = (-1A)(4V) = -\underline{4W} \text{ (generated)}$$

↑ voltage, current disagree with passive sign convention



$$P = (10V)(3A) = \underline{30W} \text{ (absorbed)}$$

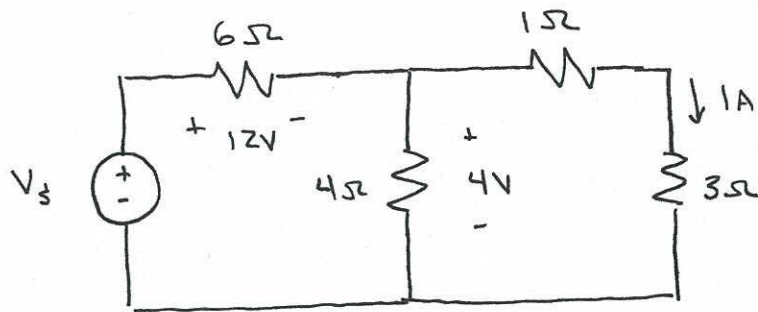
↑ Agree with passive sign convention.



$$P = (2V)(-2A) = -\underline{4W} \text{ (generated)}$$

↑ voltage, current directions agree with passive sign convention, BUT current is negative.

1.8



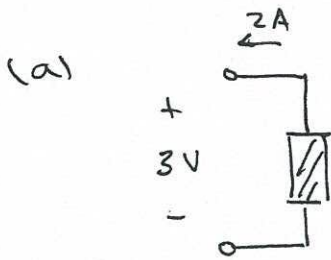
$$(a) \quad P = \frac{v^2}{R} = \frac{(12)^2}{6} = \underline{\underline{72 \text{ W}}}$$

$$(b) \quad i = 1 \text{ A} \Rightarrow v = iR \Rightarrow v = (1 \text{ A})(3 \Omega)$$

$$\underline{\underline{v = 3 \text{ V}}}$$

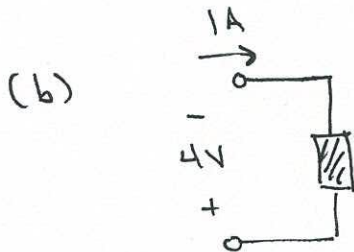
$$(c) \quad i = \frac{v}{R} \Rightarrow i = \frac{4 \text{ V}}{4 \Omega} \Rightarrow \underline{\underline{i = 1 \text{ A}}}$$

1.9



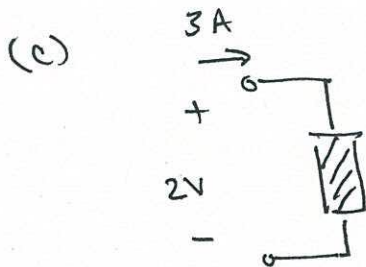
$$P = \underbrace{(-2A)(3V)}_{\uparrow} = -\underline{\underline{6W}} \text{ (generated)}$$

current, voltage don't agree with passive sign convention



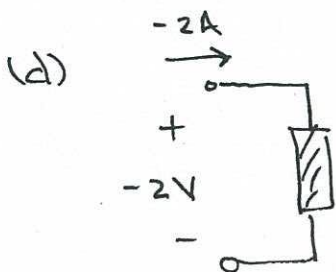
$$P = \underbrace{(-1A)(4V)}_{\uparrow} = -\underline{\underline{4W}} \text{ (generated)}$$

passive sign convention opposite to given voltage, current directions



$$P = \underbrace{(2V)(3A)}_{\uparrow} = \underline{\underline{6W}} \text{ (absorbed)}$$

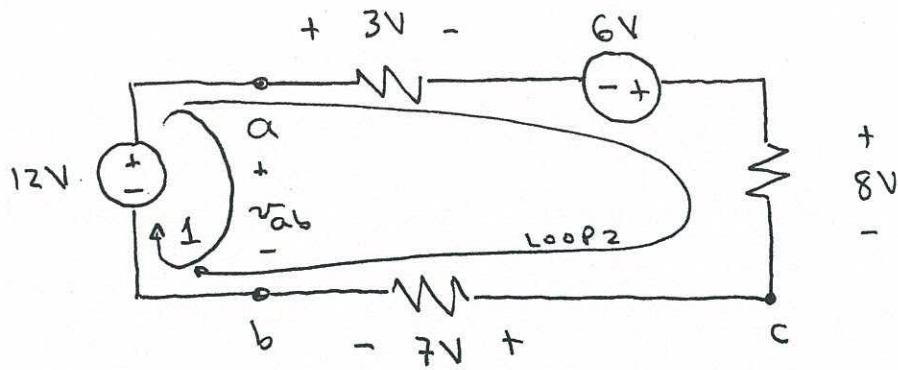
↑ voltage, current agree with passive sign convention



$$P = \underbrace{(-2A)(-2V)}_{\uparrow} = 4W \text{ (absorbed)}$$

↑ given voltage, current directions consistent with passive sign convention. But both are negative.

1.10



(a) KVL, Loop 1:

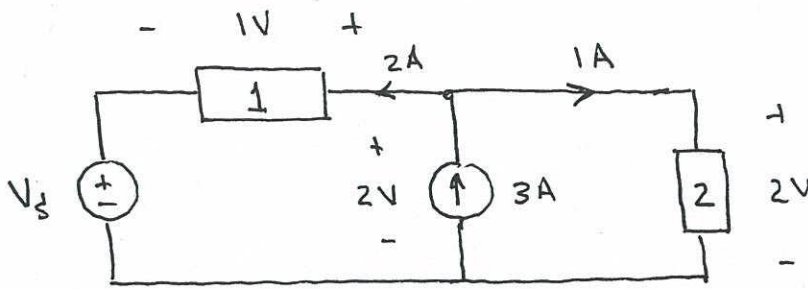
$$-12V + v_{ab} = 0 \Rightarrow \underline{\underline{v_{ab} = 12V}}$$

(b) KVL, Loop 2:

$$-12V + v_{ac} + 7V = 0 \Rightarrow \underline{\underline{v_{ac} = 5V}}$$

(c) By inspection, $\underline{\underline{v_{cb} = 9V}}$

1.11



$$P_1 = (2A)(1V) = 2W \quad (\text{note: current enters positive node})$$

$$P_{3A} = (2V)(-3A) = -6W \quad (\text{note: current leaves positive node})$$

$$P_2 = (1A)(2V) = 2W \quad (\text{note: current enters positive node})$$

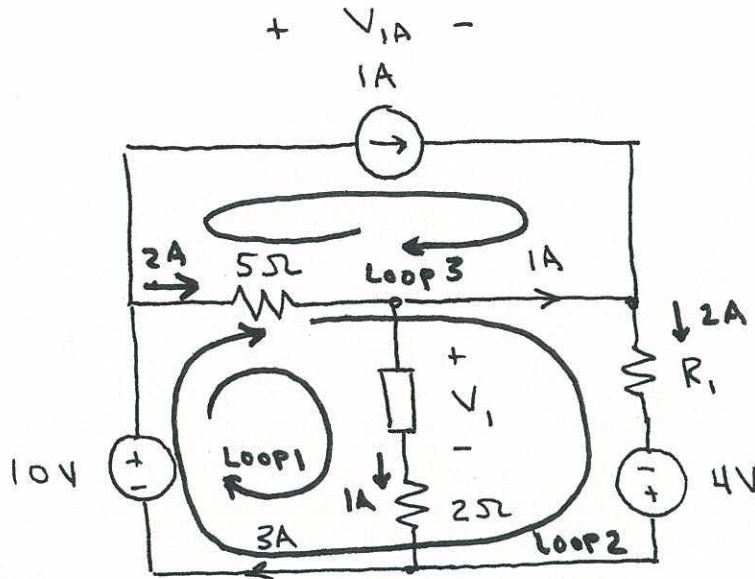
$$\sum P = 0 = P_1 + P_{3A} + P_2 + P_{V_s}$$

$$2W - 6W + 2W + P_{V_s} = 0$$

$$P_{V_s} = 2W$$

(note: the voltage source is absorbing power. This makes sense, since the 2A current enters the positive terminal of V_s .)

1.12



(a) KVL, Loop 1: $-10V + (2A)(5\Omega) + V_1 + (2\Omega)(1A) = 0$

$\uparrow$ $\uparrow$
 From KCL From KCL

$$V_1 = -2V$$

(b) KVL, Loop 2: $-10V + (2A)(5\Omega) + (2A)(R_1) - 4V = 0$ $\swarrow$ KCL

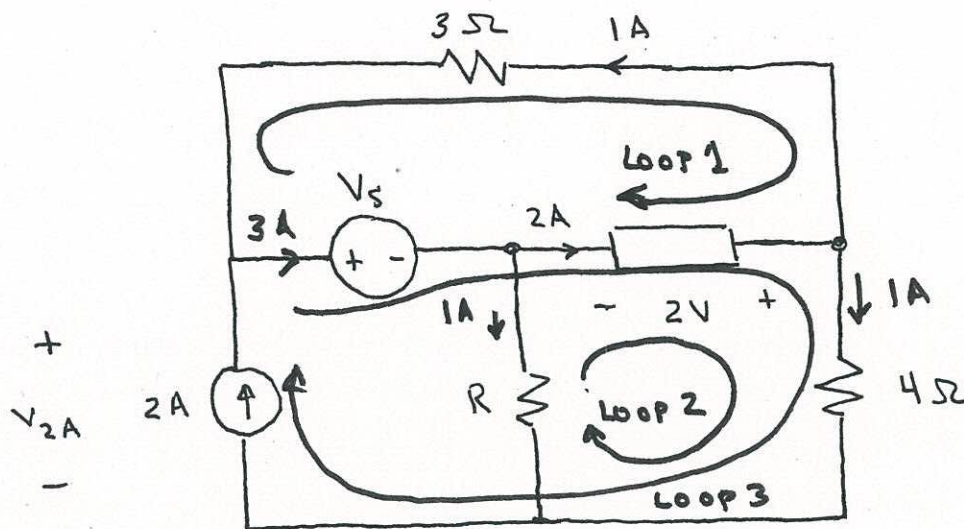
$$R_1 = 1\Omega$$

(c) KVL, Loop 3: $(-2A)(5\Omega) + V_{1A} = 0 \Rightarrow V_{1A} = 10V$

$$P = ({}_1A)({}_V{}_1A)$$

$P = 10 \text{ W}$ (absorbed; current enters positive voltage terminal)

1.13



(a) Loop 1: $-V_s - (1A)(3\Omega) + 2V = 0$

$$V_s = -1V$$

(b) Loop 2: $-R(1A) - 2V + (4\Omega)(1A) = 0 \Rightarrow R = 2\Omega$

↑
↑
 From KCL From KCL

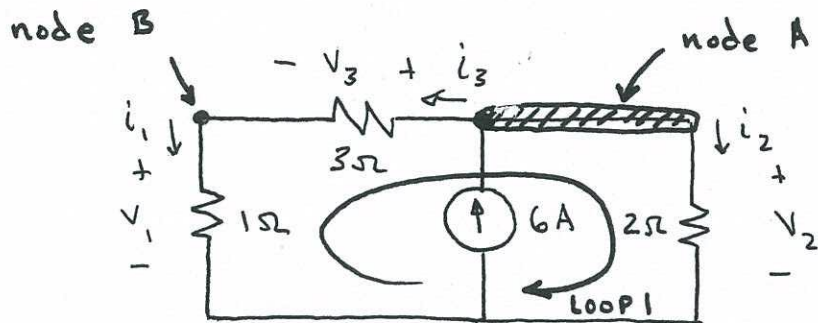
(c) Loop 3: $-V_{2A} + V_s - 2V + (4\Omega)(1A) = 0$

$$V_{2A} = 1V$$

$$P_{2A} = (1V)(-2A) = -2W \quad (\text{generated; current enters negative voltage terminal})$$

1.14

There are many different approaches, but we'll use the exhaustive approach:



$$\text{KCL, node B: } \underline{i_3 = i_1} \quad (1)$$

$$\text{KCL, node A: } \underline{6A = i_2 + i_3} \quad (2)$$

$$\text{KVL, loop 1: } \underline{-V_1 - V_3 + V_2 = 0} \quad (3)$$

Ohm's Law:

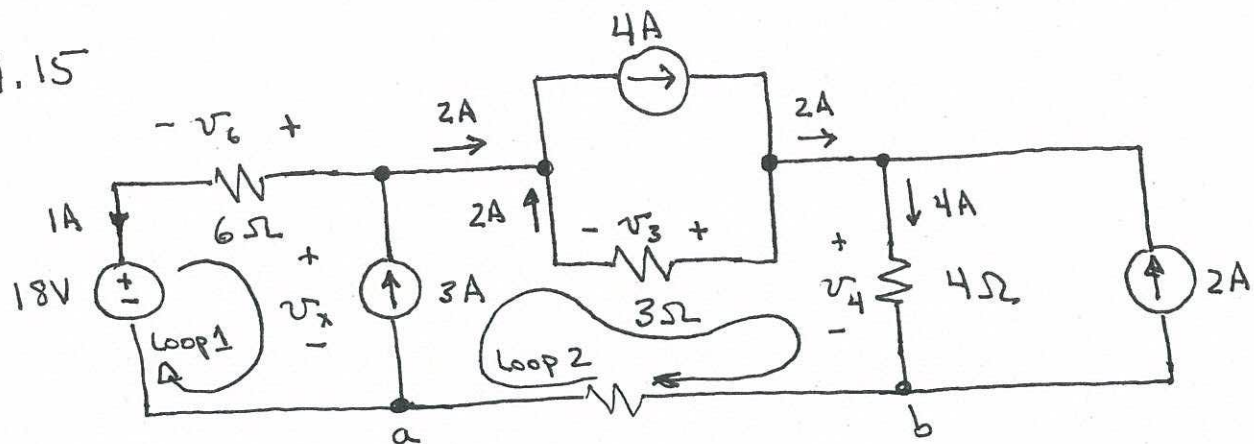
$$\underline{V_1 = (1\Omega) i_1} \quad (4)$$

$$\underline{V_2 = (2\Omega) i_2} \quad (5)$$

$$\underline{V_3 = (3\Omega) i_3} \quad (6)$$

Equations (1) - (6) comprise a system of six equations in six unknowns - we're done!

1.15



Ohm's Law: $v_6 = (6\Omega)(1A) = 6V$

KVL, Loop 1: $-18V - v_6 + v_x = 0 \Rightarrow -18V - 6V + v_x = 0$

$$v_x = 24V$$

KCL gives current through 3Ω & 4Ω resistors as shown on diagram.

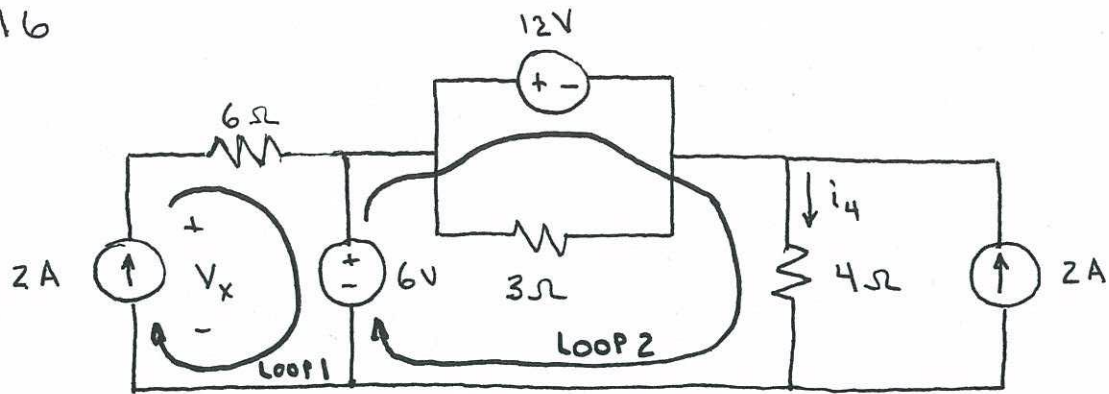
KVL, Loop 2: $-v_x - v_3 + v_4 + v_{ba} = 0$

$$-24V - (2A)(3\Omega) + (4A)(4\Omega) + v_{ba} = 0$$

$$v_{ba} = 14V \Rightarrow v_{ab} = -14V$$

$$P = iv = (-3A)(24V) = -72W \text{ (generated)}$$

1.16



$$\text{KVL, Loop 1: } -V_x + (6\Omega)(2\text{A}) + 6\text{V} = 0$$

$$\underline{\underline{V_x = 18\text{V}}}$$

$$\text{KVL, Loop 2: } -6\text{V} + 12\text{V} + (4\Omega)i_4 = 0$$

$$i_4 = \frac{-6\text{V}}{4\Omega} \Rightarrow \underline{\underline{i_4 = -1.67\text{A}}}$$